

Data-Driven Model Reduction and System Identification Bridging Learning, Approximation, and Dynamical Systems

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Abstract—Data-driven model reduction and system identification have traditionally developed as distinct yet closely related areas within dynamical systems and control. While system identification aims to construct accurate models from data, model reduction seeks to obtain low-complexity representations that retain the essential input-output behavior of dynamical systems. Recent advances increasingly blur this distinction, with data-driven techniques being used for model reduction and reduction principles being incorporated into identification methods. This open invited track aims to bring together researchers working at the intersection of these fields, with a particular focus on data-driven approaches to obtaining accurate, computationally efficient, and interpretable reduced-order models. Topics of interest include data-driven model reduction, identification of reduced-order models, structure- and physics-preserving reduction, learning-based reduction, behavioral and input-output approximation, and the integration of system identification and model reduction techniques. The track will provide a forum for discussing emerging connections between the two communities and identifying new opportunities for combining their complementary perspectives.

I. INVITED-TRACK DESCRIPTION

The increasing availability of high-dimensional experimental and simulation data has created new opportunities for combining system identification with model reduction. System identification has traditionally focused on constructing dynamical models from measured data [1]–[5], whereas model reduction seeks low-complexity representations that retain relevant dynamical or input–output properties of a given high-order model [6]–[10]. In many applications, however, identifying a high-fidelity model is only an intermediate step: the resulting model may be too complex for analysis, control design, simulation, optimization, or real-time implementation.

The emergence of black-box models further highlights this tension. Such models provide considerable flexibility for capturing complex and nonlinear system behavior directly from data [11]–[15], but their expressive power can come at the cost of increasingly complex architectures and high-dimensional representations. This raises a fundamental question: rather than first identifying a high-fidelity black-box model and subsequently reducing its complexity, can identification and model reduction be addressed jointly? Such

an approach could yield models that retain the expressive power needed to capture the observed system behavior while being sufficiently compact for subsequent analysis, control, simulation, optimization, or real-time implementation.

Conversely, model reduction methods can exploit structural, dynamical, or input–output information to obtain compact representations, but traditionally rely on a model that is already available. Recent developments increasingly remove this separation: reduced-order models can be constructed directly from input–output data through moment matching [16]–[20], obtained directly from data without explicitly identifying the full-order model [21], [22], or identified together with a reduced representation of the underlying dynamics [23], [24]. Related developments based on data-driven balancing, Koopman representations, and nonlinear manifold learning further blur the distinction between identification and reduction [25], [26]. This open invited track focuses on the intersection of these two perspectives, with the central question of how data-driven identification and model reduction can be developed jointly to obtain models that are accurate with respect to observed system behavior while remaining sufficiently low-dimensional, interpretable, and computationally efficient for subsequent use.

II. SCOPE

Contributions are invited on topics including, but not limited to:

- data-driven model order reduction and identification of reduced-order models;
- joint or integrated system identification and model reduction;
- learning-based and machine-learning approaches to model reduction;
- structure-, physics-, and stability-preserving data-driven reduction;
- identification and reduction of nonlinear, large-scale, and complex dynamical systems;
- error bounds, performance guarantees, and uncertainty quantification for data-driven reduced models;
- data-efficient approaches to identifying low-complexity dynamical models;
- identification of reduced-order representations suitable for control, prediction, and real-time applications.

The track is intended to encourage contributions from both the system identification and model reduction communities. In particular, it aims to highlight complementary methodologies, establish connections between existing approaches, and stimulate new research directions at their interface.

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III. SUBMISSION DETAILS

Details and submission instructions are available at [HERE](#).

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